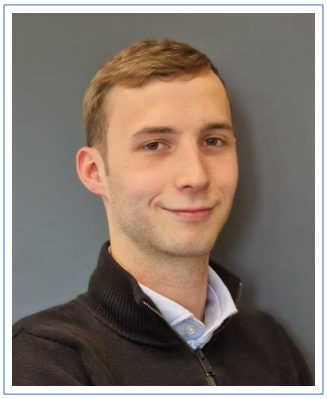
Aaron Kreis

Master's Student in Physics
M.Sc. expected 2027

Saarland University
Building E2.6, Room 3.10
66123 Saarbrücken, Germany
aakr00001@stud.uni-saarland.de
+49 1590 6170555
0009-0008-2515-2365
www.kreisaaron.de
AaronKreisvT



Experimental Physics · Optical Imaging · Scientific Computing

Research Experience

Jun–Aug 2026 **Internship – Automated Bioluminescence Microscopy and Fourier Light-Field Imaging**, *NMSB Group – Prof. Michael Krieg — ICFO – The Institute of Photonic Sciences*

- Extended and characterized a multimodal microscopy platform combining brightfield, fluorescence, bioluminescence, and Fourier light-field imaging.
- Integrated a motorized filter wheel and designed precision CAD components for optomechanical improvements.
- Developed Micro-Manager plugins for automated spiral scanning and multistep acquisition workflows, coordinating cameras, motorized stages, illumination, and filter wheels.
- Characterized the optical PSF and evaluated volumetric reconstruction using Richardson–Lucy deconvolution and deep-learning-based LUCID models.

Jul–Oct 2025 **Bachelor Thesis – Light Ray Propagation in Microscopic Imaging of Red Blood Cells**, *Dynamics of Fluids Group – Prof. Christian Wagner — Saarland University*

- Developed a numerical ray-tracing framework to simulate brightfield microscopy of red blood cells.
- Modeled light propagation through rotationally symmetric cell geometries and analyzed the resulting detector-plane intensity distributions.
- Established reproducible simulation and analysis workflows for studying image formation in microscopy.

Scientific Contributions

Journal Articles

2026 L. F. Morales Curiel, **A. Kreis**, and M. Krieg. *Seeing in the Dark: Intelligent Fourier Light Field Imaging for Bioluminescence Microscopy*. *ACS Photonics* (2026). DOI: 10.1021/acsphotonics.6c00025

Conference Posters

2026 **A. Kreis**, S. T. Hermes, T. John, and C. Wagner. *Ray-based simulation and experimental analysis of red blood cell imaging in brightfield microscopy*. DPG Spring Meeting, Dresden, Germany, poster BP 14.28.

2025 **A. Kreis**, S. T. Hermes, T. John, and C. Wagner. *Red Blood Cells under brightfield microscopy*. DPG Spring Meeting, Regensburg, Germany, poster BP 17.16.

2025 **A. Kreis** and T. John. *Brownsche Bewegung, optische Pinzetten und numerische Simulationen*. DPG School "Physikalische Praktika" (AGPP), Bad Honnef, Germany (laboratory education poster).

Technical Skills

Programming	Python, MATLAB, C++
Engineering	CAD, 3D Printing, Laser Cutting
Tools	LaTeX, Git, Linux

Awards & Grants

2025 – 2027 **Glückauf Scholarship (awarded twice)** — RAG-Stiftung

2026 **Santander Scholarship for Erasmus+ Mobility** — Santander Deutschland

2025 & 2026 **Travel Grants (awarded twice) – DPG Spring Meetings (SKM)** — WE Heraeus-Stiftung

Education

2025 – 2027 (expected) **M.Sc. Physics (Double Degree)** — Université Grenoble Alpes & Saarland University

2021 – 2025 **B.Sc. Physics** — Saarland University

2013 – 2021 **AbiBac** — Warndt-Gymnasium

Teaching Experience

2023 – 2026 **Course Instructor** — Saarland University

- Summer School "From Atoms to Zettabytes" for STEM-oriented pre-university students
- Mathematics Preparatory Course for incoming undergraduate students

2022 – 2025 **Teaching Assistant for Undergraduate Physics Courses** — Saarland University

- Led problem-solving sessions accompanying undergraduate lectures
- Guided students in experimental methods, data analysis, and scientific reporting
- Supervised laboratory courses and tutorial sessions in mechanics, electromagnetism, and computational physics

Academic Service & Leadership

2022 – present **Student Representative** — Saarland University
Examination Board (B.Sc./M.Sc. Physics); Appointment Committees; Faculty Council; Student Council

2025 – 2026 **Vice President for Partnerships** – PhITEM Student Council — Université Grenoble Alpes

2024 – 2025 **Founding Deputy Chair** – jDPG Regional Group Saarbrücken — German Physical Society

2024 **External Review Panel Member** – B.Sc. Physics Program — University of Luxembourg

Employment

2022 – 2025 **Student Assistant – Physics Outreach** — Department of Physics – Saarland University

- Designed and delivered physics demonstrations for school visits, career fairs, and public events
- Represented the department in student recruitment initiatives

2021 – 2023 **Student Assistant – Process Automation** — DEKRA Automobil GmbH

- Developed automation workflows to streamline administrative processes across multiple departments
- Reduced manual workload by optimizing repetitive tasks

Languages

German Native

English $\geq$B2

French $\geq$B2

Beyond the Lab

Warning I've been described as both brilliant and crazy. The exact ratio remains to be determined.